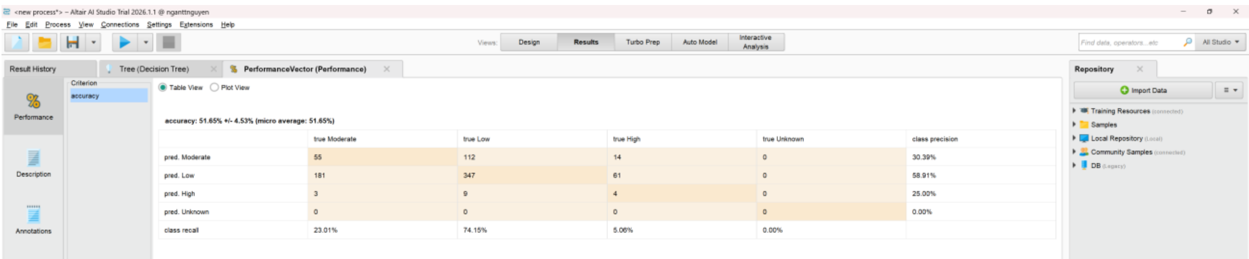
Ngan Nguyen – Lab 4 – RapidMiner

The Accuracy and Confusion Matrix (3 different trials)

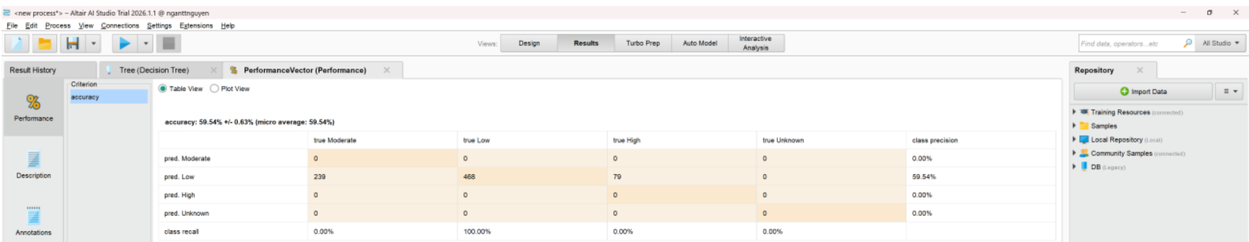
Trial 1: Accuracy = 51.65%



accuracy: 51.65% +/- 4.53% (micro average: 51.65%)

	true Moderate	true Low	true High	true Unknown	class precision
pred. Moderate	55	112	14	0	30.39%
pred. Low	181	347	61	0	58.91%
pred. High	3	9	4	0	25.00%
pred. Unknown	0	0	0	0	0.00%
class recall	23.01%	74.15%	5.06%	0.00%	

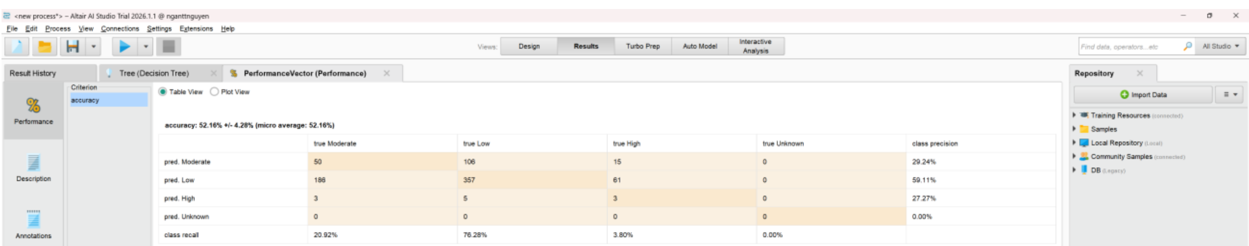
Trial 2: Accuracy = 50.34%



accuracy: 59.54% +/- 0.63% (micro average: 59.54%)

	true Moderate	true Low	true High	true Unknown	class precision
pred. Moderate	0	0	0	0	0.00%
pred. Low	239	468	79	0	59.54%
pred. High	0	0	0	0	0.00%
pred. Unknown	0	0	0	0	0.00%
class recall	0.00%	100.00%	0.00%	0.00%	

Trial 3: Accuracy = 52.16%



accuracy: 52.16% +/- 4.28% (micro average: 52.16%)

	true Moderate	true Low	true High	true Unknown	class precision
pred. Moderate	50	106	15	0	29.24%
pred. Low	186	357	61	0	59.11%
pred. High	3	5	3	0	27.27%
pred. Unknown	0	0	0	0	0.00%
class recall	20.92%	76.28%	3.80%	0.00%	

Conclusion

Trial 3 achieved the highest overall accuracy at **52.16%** by allowing the decision tree to grow deeper (maximal depth of 20) and isolate minor data pockets with a minimal leaf size of 1 or 2. Conversely, **Trial 2** performed the worst because its restrictive pre-pruning settings forced the model to oversimplify and blindly predict "Low" severity for every single accident.